

Financial Econometrics I

Project: Volatility Forecasting

Deadline: 3 days before the exam Summer semester 2025/2026 Max points: 25

General information: Form teams of two or three students. The team project must be submitted no later than three days before the oral exam. Project should be submitted to SIS in the module “Study group roster” (Studijni mezivysledky) - Financial Econometrics I - Lecture - Empirical Project as a single file for the whole team, please include names of all team members in the file name. Please use HTML generated from Jupyter notebook as a solution file. The solution should be in the form of an empirical project with all comments, discussions as well as reproducible codes. Three days before the exam, submitted projects will be distributed to other teams for peer review. Data for the homework are stored in the SIS as “data_project.zip”. “data_project.zip” contains 35 assets and each group was assigned one asset (see Table 1) - for example Group Nr. 1 is going to analyze asset number 30 stored as “30.RData”. Each “.RData” file contains an xts file with

- returns (ret),
- realized volatility (RV),
- positive realized semi-volatility (RV_p),
- negative realized semi-volatility (RV_n),
- realized skewness (RS),
- realized kurtosis (RK).

The project should include following parts:

1. **Data description:** Provide short summary of descriptive statistics, ACF plots and plots of data you are going to use.
2. **In-sample fit:** Estimate parameters of the following volatility models with full sample size: AR(1)-RV, HAR, HAR-RS, HAR-Rskew-RKurt, Realized GARCH, ARMA-GARCH. Discuss the fits and compare qualitative differences of the estimates shortly. Note you can simply use the plot comparing in-sample fits of all models.

3. **Forecasts:** Compare an out-of-sample forecasting performance of the models from previous part using two forecasting schemes:
 - (i) Expanding window, with starting window length of 750.
 - (ii) Rolling window, with the window length 750.Plot and compare forecast errors from all models, compute loss functions (MSE and MAE), compare model forecasts statistically using the Diebold-Mariano test (use the MSE loss). Evaluate all single model forecasts with the Mincer-Zarnowitz regression.
4. **Summary:** Write a short summary (up to 1/2 page) focused on comparison of model performance.

Final Exam information: The oral final exam will be based on team project presentations, peer review, and an individual oral discussion of the project and related course material. Students are allowed to use ChatGPT or similar tools when preparing the project. The assessment is not focused on the code itself, but on whether students can clearly explain what they did, what the data look like, what the results are, why the results make sense, and how the methods used in the project should be evaluated.

The expected exam format is as follows:

- Two teams will be examined together, typically three students and three students. The expected total time for the two teams is about one hour.
- Each team will present its own project in approximately 5–7 minutes.
- Each team will also review and discuss the project of the other team in approximately 5–7 minutes. The review should summarize the project and identify its main strengths and weaknesses.
- Every student must be prepared to present any part of the whole project. The examiners will decide during the exam which student presents which part.
- After the presentations and peer reviews, the remaining time will be devoted to an oral examination. The examiners will split the two teams and examine students individually within their teams. This part will focus on the project, the data, the results, interpretation, methodological choices, and relevant material from the semester.
- The exam will end with a final evaluation.

Table 1: Students Project

Group Nr.	Student 1	Student 2	Student 3	Asset
1	Babk Matj	Kuerov Tereza		30
2	Barho Tom	Jakub Hrusc	Niccolo Di Napoli	27
3	Martin Bernek	Krytof Hanzlk	Vt Vorek	29
4	Marek Pokorn	Ondej Hjek	tpn Cimler	25
5	Tom Galnor	Vojtch Strachota	Maximilin Demko	15
6	Sarah Dengov	Elika Landsmannov	Eva Lukakov	10
7	Adam Horek	Krytof Kabrhel		28
8	Alena Jechov	Tzu-Yun Lee	Dominika Ferkov	4
9	Krytof Kiml	Michal Dvoek	Jan Adam ednek	11
10	Nikola Kolnsk	Marek Malina		7
11	Hajek	Lebeda	Petrik	22
12	Kateina Ledvinov	Mubanga Sakala	Matj Knop	35
13	Mariami Leladze	Irakli Sakhvadze	Jan Zasadil	9
14	Richard Lesko	Gabriel Ferreira		31
15	Martin Liebscher	Tade Janda		14
16	Radoslav Dujsk	Dario Miku		1
17	Markta Ondejov	Pavel Bittnar	Aneta Mudrukov	20
18	Maxim Milde	Zahid Pashayev		12
19	Efa Cami	Julieta Lizarraga Rivera	Malek Sarraj	17
20	Jan Kek	Jindich Rosendorf	Michal Smieko	21
21	Zoia Solomiana	Francis Ngoma	Jasper Friis	34
22	estov	Tma	Fier	33
23	Jakub Bachorik	Vojtch Maty	Jchym vejda	5
24	Ruslan Shvets	Kemal Behnan Temiz	Narmin Vahid	19
25	Aneka Trkov	Luk Theodor Wehrenberg	Jan Trk	2
26	Daniel Ott	Samuel md	Anna Turnerov	26
27	Andrii Zatirka	Hanna Shymanska	Miloslav Bln	24
28	Krtof Pda	Marko Stamenic	Daniel Zliechovec	13
29				8
30				3
31				18
32				16
33				32
34				6
35				23